

Yuxiao (Sonia) Lai

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EMPLOYMENT

H2Evo Tech

System Analysis Engineer

Livonia, MI

Dec. 2021 - Current

EXPERIENCE

Prototype and Testing Design

Embedded System Design

Livonia, MI

Dec. 2021 - Current

- Define equipment requirements and provide technical support for implementation and integration of new equipment and systems
- Coordinate installation, testing, and training for new or updated systems
- Develop software tools to support seamless operations
- Design data acquisition kits for real-time monitoring and post-processing, including value plotting, outlier detection, and overall performance calculations
- Learn new control algorithms, software, and hardware development tools through reading, studying, attending seminars, and participating in group discussions

System Analysis

System Evaluation, Data Processing

Livonia, MI

Dec. 2021 - Current

- Create and evaluate 1D/3D systems and parametric models to improve processes and designs
- Gather and analyze runtime data and collaborate with test engineers to validate system requirements, confirm test protocols, refine test methods, and evaluate system functionality
- Identify inefficiencies using analytical tools, simulations, and root-cause analysis
- Provide technical proposals based on modeling results to optimize system performance

Embedded Software Design

Control Algorithm Design, Embedded System Programming

Livonia, MI

Dec. 2021 - Current

- Collaborate with global, multidisciplinary teams in research, design, and manufacturing
- Participate in design reviews, prototype builds, and system evaluations for vehicle components and systems
- Derive component-level specifications and requirements to ensure system functionality and performance
- Conduct hardware modifications or integrate new control algorithms to improve system performance when requirements are not met
- Identify potential causes and effects of system malfunctions
- Develop self-diagnostic procedures for fault detection
- Generate customer-side warning messages and initiate self-calibration procedures
- Implement system shutdown protocols for unresolved faults

EDUCATION

Northwestern University

M.S. in Robotics, GPA: 4.00/4.00

Evanston, IL

Aug. 2020 - Dec. 2021

Rose-Hulman Institute of Technology

B.S. in Mechanical Engineering, GPA: 3.77/4.00

Terre Haute, IN

Aug. 2016 - May 2020

Minor in Robotics, GPA: 4.00/4.00; Computer Science, GPA: 3.70/4.00; Japanese, GPA: 3.83/4.00

Selected Courses: Calculus, Differential Equations, Rhetoric and Composition, Design for Manufacturing, Analysis & Design of Engineering Systems, Machine Component Design, Embedded Systems in Robotics, Machine Dynamics, Electrical Systems, Mechanical Systems, Control Systems, Mechatronic Systems

SKILLS

Technical: Git, Linux, SolidWorks, MATLAB, MongoDB, Firebase, PyTorch, OpenCV, Arduino, Raspberry Pi, ROS, Gazebo, MoveIt, J-Link, MyCobot, dSPACE

Programming: Java, JavaScript, HTML, CSS, Kotlin, XML, Python, C/C++

Languages: Chinese (Native), English (Fluent), Japanese (Intermediate), French (Entry Level)

ACADEMIC EXPERIENCE

Myo Gesture Recognition and Robot Arm

Evanston, IL

Program Final Project

Spring - Summer 2021

https://aonai.github.io/linked_posts/myo_gesture_recognition.html

- Learn sEMG signal acquisition, processing, and analysis to relate arm muscle motions with hand gestures
- Learn to use Self-Calibrating Asynchronous Domain Adversarial Neural Network (SCADANN) to recognize hand gestures from sEMG signals recorded in the 3DC dataset
- Verify that SCADANN can improve accuracies in classifying sEMG signals recorded in the long-term Myo dataset across days, users, and wearing locations without recalibration
- Design control scheme to allow mapping right arm joints and hand gestures with the robot joints
- Acquire, process, and analyze sEMG and IMU signals from two Myo armbands to control a PincherX 100 robot arm for grabbing and placing an object
- Participated in International Consortium on Rehabilitation Robotics (ICORR) of RehabWeek Virtual 2021 Event, held a podium and a poster presentation in front of over 80 attendees

TurtleBot SLAM

Evanston, IL

Sensing, Navigation, and Machine Learning for Robotics Course Project

Winter 2021

https://github.com/aonai/TurtleBot_SLAM

- Show a TurtleBot's odometry, slam, and real trajectories in rviz with slam and real obstacles
- Write ROS nodes to control a burger TurtleBot with a commanded twist
- Design C++ libraries to perform 2D rigid body transformations, calculate forward and inverse kinematics of a two-wheel-robot, implement Feature-Based Kalman Filter SLAM, and detect landmarks from laser scan information
- Write ROS node to simulate noises while running a burger TurtleBot in the real environment with user-specified obstacles

Fast Tower

Evanston, IL

Group Member, Embedded Systems Final Project

Nov. 2020

https://aonai.github.io/linked_posts/fast_tower

- Control a Baxter robot to stack a fixed number of cups into a tower
- Build a system with Baxter ROS Interface and ROS MoveIt to plan and execute fast and stable arm motion to grab and place cups in random or specified locations
- Use ROS Apriltag to detect cups within the workstation and generate coordinate frames in rviz
- Design a simulated environment in Gazebo for debugging before running the system on the actual robot

youBot Manipulation

Evanston, IL

Robot Manipulation Final Project

Nov. 2020 – Dec. 2020

https://github.com/aonai/youbot_manipulation

- Design Python scripts to generate trajectories for a youBot to grab and place a cube from and to user-specified locations in CoppeliaSim
- Plan and compute configuration matrices along a trajectory with feedback control

Robot Speed and Separation Monitoring Safety System Design

Terre Haute, IN

Group Member, Multidisciplinary Capstone

Aug. 2019 - May 2020

- Design a safety monitoring system to ensure that robots are operating in compliance with ISO standards
- Research best practices to track workers' real-time positions and velocities in a large workspace
- Build the monitoring system using OpenCV and depth cameras to detect workers' positions and velocities
- Design stable stands and protection mounts for cameras to ensure high-quality video